LESSON 2.7
MULTIPLYING POSITIVE FRACTIONS

LESSON INTRODUCTION

MA.6.NSO.2.2 Extend previous understanding of multiplication and division to compute products and quotients of positive fractions by positive fractions with procedural fluency.

LEARNING TARGETS

- I can fluently multiply positive fractions by positive fractions.

BENCHMARK COHERENCE

BUILDING ON

MA.5.FR.2.2 Extend previous understanding of multiplication to multiply a fraction by a fraction, including mixed numbers and fractions greater than 1, with procedural reliability.

WORKING TOWARD

MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency.

ESSENTIAL QUESTIONS

- How do you multiply fractions?
- What strategies can be used to multiply fractions?
- How can fractions greater than 1 be rewritten?

LESSON NARRATIVE

This lesson begins with a Career Profile Video Warm-Up, followed by a Refresher on multiplying fractions that spirals multiplying fractions from Unit 1 and using GCF to simplify fractions from Unit 2 Lesson 1. The Refresher leads students through multiplying fractions using the standard algorithm rather than the area model. The rest of the lesson helps students work toward procedural fluency with the standard algorithm and practice a variety of problems.

COMPONENT SUMMARY

2.7.1 WARM-UP *(~5 MIN.)*

Career profile video

2.7.2 REFRESHER *(10-15 MIN.)*

Review multiplying fractions in pairs

2.7.3 GUIDED INSTRUCTION *(5-10 MIN.)*

Step-by-step practice multiplying fractions and rewriting in simplest form using GCF

2.7.4 TRY IT *(15 MIN.)*

Practice multiplying fractions and reflecting on strategies used

2.7.5 YOUR TURN! *(10 MIN.)*

Problem set to find products of mixed numbers and fractions

2.7.6 WRAP-UP *(~5 MIN.)*

Spotlight Reflection to self-assess concepts from unit

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RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Mixed number
- Greatest common factor (GCF)
- Simplest form

MATERIALS

- Career Profile Video "Food Science"

ADDITIONAL PREPARATION

- 2.7.1 Warm-Up contains a video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to recognizing equivalent fractions.
- Students may multiply when converting from an improper fraction to a mixed number.
- Students may not distribute all parts of the whole number and the fraction when multiplying.

THINGS TO CONSIDER

- Some students may need more scaffolding to recall how to convert between mixed numbers to fractions greater than 1, or vice versa.
- Consider making connections between the numerators and denominators of the factors and how this can be used to rewrite the fraction in an equivalent form. Help students recognize equivalent forms are all correct, but the preciseness of simplifying may be required.
- Consider explicitly showing how distribution happens when multiplying mixed-number fractions by using the area model to make connections (as modeled in 2.7.2 Refresher).

LITERACY AND LANGUAGE ROUTINES

Number Talk

2.7.4 Try It – Consider using Number Talk routines to allow students to think about and attempt to solve a problem individually, then share ideas and strategies as a group. This instructional routine fosters flexibility and procedural fluency. Encourage students to be precise in their word choice and use of language when describing how they found their answer.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

2.7.2 Refresher and 2.7.3 Guided Instruction position students to review previously learned concepts of fraction multiplication, the area model, and using the GCF to rewrite fractions in simplest form. Consider pointing out how these concepts connect to this lesson's targets, including how the area model and rewriting fractions connects to algorithmic procedures for multiplying fractions.

DIFFERENTIATE INSTRUCTION

2.7.5 Your Turn! - Consider placing students in groups and assigning groups to solve certain Your Turn! problems. By differentiating which groups engage with specific problems, this section of the lesson can become a jigsaw in which students are responsible for sharing their approach and answers with others.

2.7.2 Refresher provides a visual entry point by connecting the algorithm to multiplying mixed number fractions with the area model. Consider having visual learners use area models as they make connections to how the distributive property is used.

2.7.6 Wrap-Up positions students to think about their learning through "Stoplight Reflections" for important operational concepts from this unit. Consider a variety of methods for displaying and interpreting these reflections.

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2.7.1 WARM-UP: FOOD SCIENCE

Component Overview: *Display the "Career Profile – Food Science" video for the entire class. After students watch this video, prompt the students to answer the question independently. Consider giving students an opportunity to share their responses with classmates.*

QUESTIONING

Do you know someone who is a research scientist or works in food science? Consider bringing in a visitor for your class to ask questions and learn from. Additionally, consider having students return to this lesson later in the year for project-based learning around this Career Profile.

2.7.1 Warm-Up: Food Science

Dr. Charlotte Armah is a research scientist at the Institute of Food Research. She leads experiments involving human volunteers to learn whether or not eating particular foods can protect us from diseases such as cardiovascular disease and cancer. Her work allows her to combine her scientific and math expertise, interest in helping others, and enjoyment of interacting with different people.

1. **What is one aspect of Charlotte's story that was surprising to you? Explain why this stood out to you.**

   Answers will vary.

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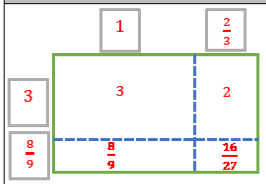
2.7.2 Refresher: Multiplying Fractions

When **multiplying fractions**, multiply the numerators of the factors to find the numerator of the product, and multiply the denominators of the factors to find the denominator of the product.

$$\frac{3}{4}\times\frac{2}{5}=\frac{6}{20}$$

1. With your partner, determine who will be partner A and who will be partner B.
   a. Find the product $1\frac{2}{3}\times 3\frac{8}{9}$ using the method indicated.

Partner A: Area Model	Partner B: Rewrite the Mixed Number(s) as a Fraction Greater than 1
 $3+2+\frac{8}{9}+\frac{16}{27}=6\frac{13}{27}$	$1\frac{2}{3}\times 3\frac{8}{9}$ $\frac{5}{3}\cdot\frac{35}{9}=\frac{175}{27}=6\frac{13}{27}$

   a. Compare your work with your partner to verify that you both found the same product. Discuss any errors and make corrections as needed. Once you are in agreement, write your partner's work in your workbook in the space provided.

2.7.2 REFRESHER: MULTIPLYING FRACTIONS

Component Overview: *Students work in partners to review how to multiply fractions. This section helps connect the algorithmic approach with the area model and with rewriting mixed numbers as a fraction greater than 1 before multiplying. Though the area model is used as a scaffold and to make connections, be sure to reinforce procedural steps using the standard algorithm to achieve algorithmic fluency.*

ENRICHMENT

Consider asking students the following:

- Are 6 13/27 and 175/27 equivalent?
- Do products found as fractions greater than 1 always need to be rewritten as mixed numbers?

TEACHER MOVES

After students have time to compare strategies, ask if the same product could be attained by multiplying the whole number parts by each other and the fractional parts by each other and then combining. Ensure students understand the distribution that is happening and which parts are being multiplied using the area model to make connections.

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2.7.3 GUIDED INSTRUCTION: MULTIPLYING FRACTIONS

Component Overview: *This section guides students through multiplying fractions step-by-step, including first rewriting mixed numbers and multiplying, then using the GCF of the numerator and denominator to write an equivalent fraction. Consider completing the first question together as a class or in partners to review how to use the GCF to rewrite fractions in simplest form from Lesson 1.*

TEACHER MOVES

Encourage recall of concepts and math vocabulary from Lesson 1 prior to starting the problems in this section. Questions could include: “What is the GCF? How do we find the GCF? How is the GCF used to write a fraction in an equivalent form?”

DIFFERENTIATED INSTRUCTION

To strengthen understanding for visual learners, consider having students draw arrows from each step in the left column to their work on the right column. This can help students make sure they completed each step, connect a written step to a visual example, and encourage procedural understanding on how to multiply fractions step-by-step.

ENRICHMENT

For question 2, ask students, “How can you determine that the product will be able to be simplified when looking at the factors multiplied?” Students may offer multiple methods, some of which do not include the greatest common factor. Build fluency (MA.K12.MTR.3.1) by emphasizing the usefulness of the greatest common factor.

2.7.3 Guided Instruction: Multiplying Fractions

Recall from Lesson 1 that fractions can be rewritten in simplest form using the GCF of the numerator and denominator.

Find each product using the steps indicated.

1.

$$\frac{5}{8} \times 4\frac{2}{3}$$

Write the given multiplication expression.	$\frac{5}{8} \cdot 4\frac{2}{3}$
Rewrite the expression, rewriting any mixed numbers as fractions greater than 1.	$\frac{5}{8} \cdot \frac{14}{3}$
Multiply to find the product of the fractions.	$\frac{70}{24}$
Rewrite the product in simplest form using the GCF of the numerator and denominator.	$\frac{35}{12}$ *or* $2\frac{11}{12}$

2.

$$3\frac{3}{5} \cdot 2\frac{1}{6}$$

Write the given multiplication expression.	$3\frac{3}{5} \cdot 2\frac{1}{6}$
Rewrite the expression, rewriting any mixed numbers as fractions greater than 1.	$\frac{18}{5} \cdot \frac{13}{6}$
Multiply to find the product of the fractions.	$\frac{234}{30}$
Rewrite the product in simplest form using the GCF of the numerator and denominator.	$\frac{39}{5}$ *or* $7\frac{4}{5}$

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2.7.4 Try It: Multiplying Fractions

1. Find each product. Rewrite each product in simplest form. Show your work.

a. $\left(1\frac{1}{8}\right)\left(5\frac{5}{7}\right)$

$\frac{9}{8} \cdot \frac{40}{7} = \frac{360}{56} = \frac{45}{7}$ *or* $6\frac{3}{7}$

b. $4\frac{4}{5} \times \frac{10}{13}$

$\frac{24}{5} \cdot \frac{10}{13} = \frac{240}{65} = \frac{48}{13}$ *or* $3\frac{9}{13}$

c. $\frac{4}{11} \cdot \frac{1}{2}$

$\frac{4}{11} \cdot \frac{1}{2} = \frac{4}{22} = \frac{2}{11}$

d. $2\frac{1}{3} \times 4\frac{4}{5}$

$\frac{7}{3} \cdot \frac{24}{5} = \frac{168}{15} = \frac{56}{5}$ *or* $11\frac{1}{5}$

2. Complete the statement to describe which product was the most challenging for you to find.
Answers will vary.

The product of __________ was most challenging for me

to find because ____

____.

3. What is a strategy that could help you on a similar problem in the future?
Answers will vary.

2.7.4 TRY IT: MULTIPLYING FRACTIONS

Component Overview: *Students have four questions to practice multiplying fractions and rewrite in simplest form. Consider using any part of question 1 for Number Talk routines and practicing procedural fluency, then question 2 to position students to reflect on which questions were most challenging and on preferred strategies for future problems.*

TEACHER MOVES

Number Talk

Consider using any of the questions 1a-d as an opportunity to engage in "Number Talk" routines.

Pick one problem for students to consider different approaches to solving, then share those ideas as a group.

Highlight responses that:

- Rewrite mixed numbers to simplify or reduce before multiplying (fluency).
- Utilize both GCF and other common factors.
- Utilize precision in math vocabulary.
- Show different "correct" answers to compare more precise responses, like 4/22 or 2/11 in 1c.

2.7.5 Your Turn!

1. Find each product. Simplify when possible.

a. $\frac{2}{3} \times \frac{5}{4}$

$\frac{10}{12} = \frac{5}{6}$

b. $2\frac{1}{3} \times \frac{3}{4}$

$\frac{7}{3} \cdot \frac{3}{4} = \frac{21}{12} =$

$\frac{7}{4}$ *or* $1\frac{3}{4}$

c. $\frac{8}{7} \cdot \frac{7}{10}$

$\frac{56}{70} = \frac{4}{5}$

d. $\frac{13}{2} \cdot \frac{7}{4}$

$\frac{91}{8}$ *or* $11\frac{3}{8}$

e. $7\frac{5}{6} \times 9$

$\frac{47}{6} \cdot \frac{9}{1} = \frac{423}{6} =$

$\frac{141}{2}$ *or* $70\frac{1}{2}$

f. $\left(\frac{24}{27}\right)\left(\frac{45}{32}\right)$

$\frac{8}{9} \cdot \frac{9 \cdot 5}{8 \cdot 4} = \frac{5}{4}$

g. $\frac{3}{2} \cdot \frac{7}{10}$

$\frac{21}{20}$ *or* $1\frac{1}{20}$

h. $\left(2\frac{6}{25}\right)\left(\frac{1}{25}\right)$

$\frac{56}{25} \cdot \frac{1}{25} = \frac{56}{625}$

i. $4\frac{1}{5} \times 1\frac{7}{9}$

$\frac{21}{5} \cdot \frac{16}{9} = \frac{336}{45} =$

$\frac{112}{15}$ *or* $7\frac{7}{15}$

2.7.5 YOUR TURN!

Component Overview: *Students independently practice finding products of mixed numbers and fractions and simplifying when possible. Consider having students work in groups to solve three problems, then share answers and strategies in a jigsaw activity.*

DIFFERENTIATED INSTRUCTION

Students should independently engage with problems for meaningful practice.

If more collaboration is necessary depending on the level of your students, consider splitting the class into groups and assigning each group three problems to complete, then share how they solved each problem with the other two groups. Note that questions 1b, 1e, and 1h include mixed numbers, and all students should independently engage with at least two of these questions.

This jigsaw activity will provide an opportunity for students to share strategies and work on precise and accurate math language as they describe their strategy and share their answers with the other groups.

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2.7.6 WRAP-UP: REFLECTION

Component Overview: *Students use the "Stoplight Reflection" format to self-assess learning on important concepts from this unit. Consider having students write responses, share how they are feeling verbally to a partner, or share their reflection with the teacher individually.*

DIFFERENTIATED INSTRUCTION

Notice the inclusion of concept stems for students to use as additional entry points for engaging in self-reflection.

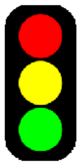
- Consider providing an example of each concept for students to use as an anchor for reflection.
- Consider providing each student with four mini-flag sticky notes for each concept and have chart paper displayed with the stoplight. This visual representation will give you an aggregate of all students' reflections.

2.7.6 Wrap-Up: Reflection

In this unit, you have learned multiple concepts including those listed below.

- Finding the GCF and LCM of two or more numbers
- Rewriting sums using distributive property
- Multiplying and dividing decimals
- Multiplying fractions

1. Use the stoplight reflection below to share how you are feeling about each of these concepts at this point.
   Answers will vary.



I still feel stuck about how to...

I'm starting to get the hang of...

I feel confident and ready to go with...

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